Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 1 of 7 · Question 6 part c · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| **6(a)** | short term is recent event/long term goes back many years;<br>short term very little processing, long term has more processing; | | **(2)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(b)** | cerebral hemispheres/cerebral cortex/frontal lobe/cerebrum; | | **(1)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(c)** | • language/speaking/talking problems / repetitive actions/confusion;<br>• loss of voluntary muscle function;<br>• loss of intellectual capacity/function;<br>• deterioration in social behaviour; | | **(2 max)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(d)** | • blocks synapse/cleft/build-up of plaques;<br>• prevents diffusion of acetylcholine / transmitter/neurones communicating;<br>• blocks post-synaptic receptors/prevents generation of new impulse;<br>• death of neurones; | | **(3 max)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **6(e)** | • inhibits/prevents breakdown of acetylcholine / transmitter;<br>• increases chance of synaptic transmission / neurones communicating/generating new impulse;<br>• reduces death of neurones; | | **(2)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(f)** | formation of β-amyloid is caused by incorrect breakdown of APP;<br>build-up/formation of β-amyloid (in neurones); | | **(2)** |

**Total 12 Marks**

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 1 of 7 · Question 6 part c · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

**Paper 2: Subject Human Biology**

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 2 of 7 · Question 6 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

| | | | |
|---|---|---|---|
| **6(a)** | short term is recent event/long term goes back many years;<br>short term very little processing, long term has more processing; | | **(2)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(b)** | cerebral hemispheres/cerebral cortex/frontal lobe/cerebrum; | | **(1)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(c)** | • language/speaking/talking problems / repetitive actions/confusion;<br>• loss of voluntary muscle function;<br>• loss of intellectual capacity/function;<br>• deterioration in social behaviour; | | **(2 max)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(d)** | • blocks synapse/cleft/build-up of plaques;<br>• prevents diffusion of acetylcholine / transmitter/neurones communicating;<br>• blocks post-synaptic receptors/prevents generation of new impulse;<br>• death of neurones; | | **(3 max)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **6(e)** | • inhibits/prevents breakdown of acetylcholine / transmitter;<br>• increases chance of synaptic transmission / neurones communicating/generating new impulse;<br>• reduces death of neurones; | | **(2)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(f)** | formation of β-amyloid is caused by incorrect breakdown of APP;<br>build-up/formation of β-amyloid (in neurones); | | **(2)** |

**Total 12 Marks**

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 2 of 7 · Question 6 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

**Paper 2: Subject Human Biology**

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 3 of 7 · Question 6 part f · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part f in this mark scheme extract

| | | | |
|---|---|---|---|
| **6(a)** | short term is recent event/long term goes back many years;<br>short term very little processing, long term has more processing; | | **(2)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(b)** | cerebral hemispheres/cerebral cortex/frontal lobe/cerebrum; | | **(1)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(c)** | • language/speaking/talking problems / repetitive actions/confusion;<br>• loss of voluntary muscle function;<br>• loss of intellectual capacity/function;<br>• deterioration in social behaviour; | | **(2 max)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(d)** | • blocks synapse/cleft/build-up of plaques;<br>• prevents diffusion of acetylcholine / transmitter/neurones communicating;<br>• blocks post-synaptic receptors/prevents generation of new impulse;<br>• death of neurones; | | **(3 max)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **6(e)** | • inhibits/prevents breakdown of acetylcholine / transmitter;<br>• increases chance of synaptic transmission / neurones communicating/generating new impulse;<br>• reduces death of neurones; | | **(2)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(f)** | formation of β-amyloid is caused by incorrect breakdown of APP;<br>build-up/formation of β-amyloid (in neurones); | | **(2)** |

**Total 12 Marks**

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 3 of 7 · Question 6 part f · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part f in this mark scheme extract

**Paper 2: Subject Human Biology**

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 4 of 7 · Question 5 part b.i · 2 marks

2023 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 5 | (a) (i) | Any two from:<br>• chemical;<br>• diffused from one neurone to another;<br>• allows neurones to communicate / reference to communication between presynaptic and post synaptic neurones; | | 2 |
| | (ii) | • neurotransmitter/dopamine/ diffuses/travels across synapse;<br>• binds with receptor (on postsynaptic membrane) | | 2 |
| | (iii) | Any two from:<br>• increased movement;<br>• increased heart rate/blood pressure;<br>• more energy;<br>• sleeping difficulties;<br>• anxiety/stress;<br>• hallucinations; | | 2 |
| | (b) (i) | Any two from<br>• tremors/random muscular movement / involuntary movement;<br>• slower movement;<br>• speech/writing changes;<br>• impaired posture/balance;<br>• sleeping problems;<br>• rigidity/stiffness;<br>• anxiety/depression/mental health/memory problems; | | 2 |
| | (ii) | • (145 000 ÷ 100) (x 18) = 26 100;<br>• (26 100 + 145 000) = 171 100; | Allow ecf for mp2 - if mp1 has an incorrect value but this value is added to 145 000 to arrive at the correct answer for the calculation award 1 mark<br><br>Full marks for correct final answer | 2 |

**Total for question 5 = 10 marks**

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 5 of 7 · Question 4 part c.iv · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part c.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 4 (a) | • diarrhoea by clean water/safe water/ bottled water/boiled water/ sewage treatment/chlorination/good sanitation/not mixing sewage with drinking water/washing hands/good hygiene/ cooking food thoroughly;<br>• malaria by nets/draining swamps/insecticides/ insect repellent's/wearing clothing to cover arms (and legs/mosquito coils/anti-malarial/vaccination ; | Ignore ORT | 2 |
| (b) (i) | antibiotics; | | 1 |
| (ii) | • elderly have weakened/lowered immune system/immunocompromised;<br>• indoor spaces makes transmission easier/reduced air flow/ reduced ventilation/build- up of air droplets with pathogens/ people closer together; | | 2 |
| (c) (i) | • diet;<br>• exercise;<br>• smoking:<br>• drinking alcohol:<br>• stress | Accept drugs | 2 |
| (ii) | • $\frac{5\,000\,000}{100\,000} = 50$;<br>• $50 \times 445 = 22\,250$;<br>• 22 300;<br>***22,250 scores 2 marks***<br>***22,300 scores 3 marks*** | ***1 mark for ×445 if there are no other marks*** | 3 |
| (iii) | • use of 445 and 185;<br>• $\frac{445}{185}$ ;<br>• 2.41:1/2.405:!/2.4:1= 3 marks; | ecf from mp1<br>**if they have done the correct calculation and 2:1 is on the answer line they score** 2 marks | 3 |
| (iv) | • people live longer/elderly population;<br>• these diseases are more common in older people;<br>• less likely to die younger from infectious diseases/better medical facilities/treatment;<br>• different diet/ more meat/saturated fats/processed food;<br>• drinking more alcohol; | People live longer so there is more cancer and dementia/these diseases= 2 marks | 2 |

**Total 15 marks**

## Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

### Answer 5 of 7 · Question 4 part c.iv · 2 marks

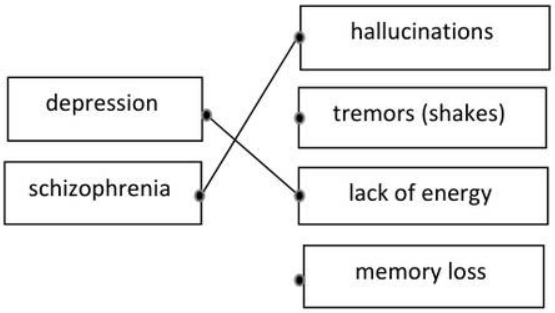
2024 NOV · Paper 02 · extract 2 of 2 · find part c.iv in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 6 of 7 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part a.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) | 2002; | | 1 |
| (ii) | 2005 (1.4) | | 1 |
| (iii) | 2.75:2.6;<br>1.1/1.06 :1; | | 2 |
| (iv) | low income/low self esteem/(domestic) violence/low social status/substance abuse; | | 1 |
| (b) |  | | 2 |
| (c) | **A** alcohol and heroin;<br>B antibiotics have no effect on mental health<br>C paracetamol has no effect on mental health<br>D antibiotics have no effect on mental health | | 1 |
| (d) | Any two from the following<br>brain/skin/sweat glands/muscles/blood vessels/arterioles;; | | 2 |
| | | | Total 10 |

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 6 of 7 · Question 2 part a.iv · 1 mark

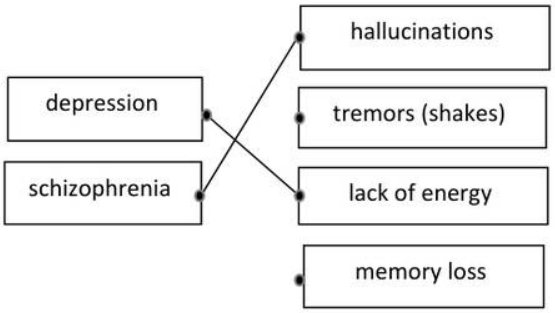
2022 JUN · Paper 02 · extract 2 of 2 · find part a.iv in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 7 of 7 · Question 2 part b · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) | 2002; | | 1 |
| (ii) | 2005 (1.4) | | 1 |
| (iii) | 2.75:2.6;<br>1.1/1.06 :1; | | 2 |
| (iv) | low income/low self esteem/(domestic) violence/low social status/substance abuse; | | 1 |
| (b) |  | | 2 |
| (c) | **A** alcohol and heroin;<br><br>B antibiotics have no effect on mental health<br>C paracetamol has no effect on mental health<br>D antibiotics have no effect on mental health | | 1 |
| (d) | Any two from the following<br><br>brain/skin/sweat glands/muscles/blood vessels/arterioles;; | | 2 |
| | | | Total 10 |

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 7 of 7 · Question 2 part b · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|